

com.holidaypackage.client

Main.java

```
package com.holidaypackage.client;
import java.util.List;
import java.util.Scanner;
import com.holidaypackage.dao.InvalidPackageIdException;
import com.holidaypackage.model.Package;
import com.holidaypackage.service.PackageService;
import com.holidaypackage.service.PackageServiceImpl;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        PackageService service = new PackageServiceImpl();
        while (true) {
            System.out.println("\n1. Add Package Details");
            System.out.println("2. Display All Packages");
            System.out.println("3. Search Package by ID");
            System.out.println("4. Calculate Package Cost by ID");
            System.out.println("5. Exit");
            System.out.print("Enter your choice: ");
            int choice = sc.nextInt();
            sc.nextLine();
            switch (choice) {
                case 1:
                    try {
                        System.out.print("Enter Package ID (7 chars): ");
                        String id = sc.nextLine();
                        System.out.print("Enter Source Place: ");
                        String source = sc.nextLine();
                        System.out.print("Enter Destination Place: ");
                        String dest = sc.nextLine();
                        System.out.print("Enter No. of Days: ");
                        int days = sc.nextInt();
                        System.out.print("Enter Basic Fare: ");
                        double fare = sc.nextDouble();
                        sc.nextLine();
                        Package pack = new Package(id, source, dest, days, fare);
                        service.addPackage(pack);
                        System.out.println("Package added successfully.");
                    } catch (InvalidPackageIdException e) {
                        System.out.println(e.getMessage());
                    }
                    break;
                case 2:
                    List<Package> packages = service.fetchAllPackages();
                    if (packages.isEmpty()) {
```

```

        System.out.println("No packages available.");
    } else {
        for (Package p : packages) {
            System.out.println(p);
        }
    }
    break;
case 3:
    System.out.print("Enter Package ID to search: ");
    String searchId = sc.nextLine();
    Package result = service.findPackageById(searchId);
    if (result != null) {
        System.out.println(result);
    } else {
        System.out.println("Package not found.");
    }
    break;
case 4:
    try {
        System.out.print("Enter Package ID to calculate cost: ");
        String costId = sc.nextLine();
        service.calculatePackageCost(costId);
        System.out.println("Package cost calculated.");
        Package updated = service.findPackageById(costId);
        System.out.println(updated);
    } catch (InvalidPackageIdException e) {
        System.out.println(e.getMessage());
    }
    break;
case 5:
    System.out.println("Exiting...");
    sc.close();
    return;
default:
    System.out.println("Invalid choice.");
}
}
}
}

```

com.holidaypackage.model

Package.java

```

package com.holidaypackage.model;
public class Package {
    private String packageId;
    private String sourcePlace;

```

```
private String destinationPlace;
private int noOfDays;
private double basicFare;
private double packageCost;
public Package(String packageld, String sourcePlace, String destinationPlace, int noOfDays, double
basicFare) {
    this.packageld = packageld;
    this.sourcePlace = sourcePlace;
    this.destinationPlace = destinationPlace;
    this.noOfDays = noOfDays;
    this.basicFare = basicFare;
}
public String getPackageld() {
    return packageld;
}
public void setPackageld(String packageld) {
    this.packageld = packageld;
}
public String getSourcePlace() {
    return sourcePlace;
}
public void setSourcePlace(String sourcePlace) {
    this.sourcePlace = sourcePlace;
}
public String getDestinationPlace() {
    return destinationPlace;
}
public void setDestinationPlace(String destinationPlace) {
    this.destinationPlace = destinationPlace;
}
public int getNoOfDays() {
    return noOfDays;
}
public void setNoOfDays(int noOfDays) {
    this.noOfDays = noOfDays;
}
public double getBasicFare() {
    return basicFare;
}
public void setBasicFare(double basicFare) {
    this.basicFare = basicFare;
}
public double getPackageCost() {
    return packageCost;
}
public void setPackageCost(double packageCost) {
    this.packageCost = packageCost;
}
@Override
```

```

public String toString() {
    return "PackageId: " + packageId + ", Source: " + sourcePlace +
        ", Destination: " + destinationPlace + ", Days: " + noOfDays +
        ", Basic Fare: " + basicFare + ", Package Cost: " + packageCost;
}
}

```

Com.holidaypackage.dao

PackageDao.java

```

package com.holidaypackage.dao;
import java.util.List;
import com.holidaypackage.model.Package;
public interface PackageDao {
    void addPackage(Package pack);
    List<Package> getAllPackages();
    Package getPackageById(String packageId);
}

```

PackageDaoImpl.java

```

package com.holidaypackage.dao;
import java.util.ArrayList;
import java.util.List;
import com.holidaypackage.model.Package;
public class PackageDaoImpl implements PackageDao {
    private List<Package> packageList = new ArrayList<>();
    @Override
    public void addPackage(Package pack) {
        packageList.add(pack);
    }
    @Override
    public List<Package> getAllPackages() {
        return packageList;
    }
    @Override
    public Package getPackageById(String packageId) {
        for (Package p : packageList) {
            if (p.getPackageId().equals(packageId)) {
                return p;
            }
        }
        return null;
    }
}

```

InvalidPackageldException.java

```
package com.holidaypackage.dao;
public class InvalidPackageldException extends Exception {
    public InvalidPackageldException(String message) {
        super(message);
    }
}
```

Com.holidaypackage.service

PackageService.java

```
package com.holidaypackage.service;
import java.util.List;
import com.holidaypackage.dao.InvalidPackageldException;
import com.holidaypackage.model.Package;
public interface PackageService {
    void addPackage(Package pack) throws InvalidPackageldException;
    List<Package> fetchAllPackages();
    Package findPackageById(String packageld);
    void calculatePackageCost(String packageld) throws InvalidPackageldException;
}
```

PackageServiceImpl.java

```
package com.holidaypackage.service;
import java.util.List;
import com.holidaypackage.dao.InvalidPackageldException;
import com.holidaypackage.dao.PackageDao;
import com.holidaypackage.dao.PackageDaoImpl;
import com.holidaypackage.model.Package;
public class PackageServiceImpl implements PackageService {
    private PackageDao dao = new PackageDaoImpl();
    @Override
    public void addPackage(Package pack) throws InvalidPackageldException {
        if (!isValidPackageld(pack.getPackageld())) {
            throw new InvalidPackageldException("Invalid Package Id");
        }
        dao.addPackage(pack);
    }
    @Override
    public List<Package> fetchAllPackages() {
        return dao.getAllPackages();
    }
    @Override
    public Package findPackageById(String packageld) {
        return dao.getPackageById(packageld);
    }
}
```

```

}
@Override
public void calculatePackageCost(String packageld) throws InvalidPackageIdException {
    Package pack = dao.getPackageById(packageld);
    if (pack == null) {
        throw new InvalidPackageIdException("Invalid Package Id");
    }
    double basicTotal = pack.getBasicFare() * pack.getNoOfDays();
    double discount = 0;
    int days = pack.getNoOfDays();
    if (days > 5 && days <= 8) {
        discount = basicTotal * 0.03;
    } else if (days > 8 && days <= 10) {
        discount = basicTotal * 0.05;
    } else if (days > 10) {
        discount = basicTotal * 0.07;
    }
    double costAfterDiscount = basicTotal - discount;
    double gst = costAfterDiscount * 0.12;
    double totalCost = costAfterDiscount + gst;
    pack.setPackageCost(totalCost);
}
private boolean isValidPackageId(String packageld) {
    return packageld != null && packageld.length() == 7;
}
}

```

Output:

1. Add Package Details
2. Display All Packages
3. Search Package by ID
4. Calculate Package Cost by ID
5. Exit

Enter your choice: 1

Enter Package ID (7 chars): pack001

Enter Source Place: erode

Enter Destination Place: goa

Enter No. of Days: 10

Enter Basic Fare: 40000

Package added successfully.

-
1. Add Package Details
 2. Display All Packages
 3. Search Package by ID
 4. Calculate Package Cost by ID

5. Exit

Enter your choice: 1

Enter Package ID (7 chars): pack002

Enter Source Place: tirupur

Enter Destination Place: chennai

Enter No. of Days: 6

Enter Basic Fare: 30000

Package added successfully.

1. Add Package Details

2. Display All Packages

3. Search Package by ID

4. Calculate Package Cost by ID

5. Exit

Enter your choice: 2

Packageld: pack001, Source: erode, Destination: goa, Days: 10, Basic Fare: 40000.0, Package Cost: 0.0

Packageld: pack002, Source: tirupur, Destination: chennai, Days: 6, Basic Fare: 30000.0, Package Cost: 0.0

1. Add Package Details

2. Display All Packages

3. Search Package by ID

4. Calculate Package Cost by ID

5. Exit

Enter your choice: 3

Enter Package ID to search: pack002

Packageld: pack002, Source: tirupur, Destination: chennai, Days: 6, Basic Fare: 30000.0, Package Cost: 0.0

1. Add Package Details

2. Display All Packages

3. Search Package by ID

4. Calculate Package Cost by ID

5. Exit

Enter your choice: 4

Enter Package ID to calculate cost: pack001

Package cost calculated.

Packageld: pack001, Source: erode, Destination: goa, Days: 10, Basic Fare: 40000.0, Package Cost: 425600.0
